NovaRetail
EUROPEAN RETAIL · DATA · GENAI COPILOT

RETAIL ANALYTICS CASE STUDY

The Hidden Cost of Aggressive Discounting in Fashion

A margin and customer experience problem concentrated in the UK & Ireland in the fourth quarter, and the AI response best suited to solve it

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Data source: NovaRetail_Analytical_Copilot_Dataset_v3.xlsx · Analysis in Python (pandas/matplotlib) · GenAI as analytical copilot

Independent case study built for a Generative AI applications course, using a fictional retail dataset.

Executive summary

This report identifies a concrete business problem at NovaRetail, drawn from the case study and the 2025 dataset. In Q4 (October to December), the Fashion category in the UK & Ireland showed an isolated deterioration: average discount rose from 7.3% to 17.5%, returns from 8.3% to 11.0%, and gross margin fell from 53.9% to 48.1%. By contrast, Italy kept discount stable (6.9% to 6.6%) and even improved returns and margin over the same period. Operational indicators stayed stable in both regions, which rules out a logistics cause.

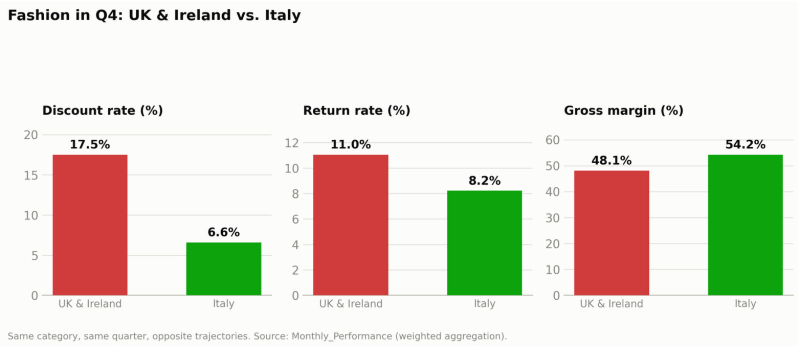

Figure A. Fashion in Q4: UK & Ireland vs. Italy, the two regions that best illustrate the problem.

Using Generative AI as an analytical copilot, always with programmatic verification in Python, commercial, campaign, operations, customer service and feedback data were cross-referenced to separate supported observation, plausible hypothesis and points still to validate. The three Fashion campaigns in the UK & Ireland in Q4 have ROAS below break-even (0.31x to 0.74x, with no incrementality test), and CSAT falls from 78.8 to 72.0. The recommendation is therefore to equip regional and category managers with a Generative AI interface that lets them explore these signals through structured, auditable explanations.

1. Problem framing

NovaRetail is a European retail group with six management regions (DACH, France, Iberia, Italy, UK & Ireland, Nordics), six categories and three sales channels. In 2025 it generated €80.5 million in gross sales, with an average gross margin of 34.5% and an overall return rate of 5.5%. The executive team suspects there is no single, company-wide problem, but rather distinct combinations of problems by region and category, and that is the hypothesis the data confirms.

The analysis below shows that the risk of sales won at the expense of returns and margin is the dataset's strongest and most isolable pattern: it is concentrated in Fashion, UK & Ireland, Q4. It matters because Q4 is the most commercially important quarter of the year, because it does not stem from general seasonality, and because the same signal appears across four independent data sources.

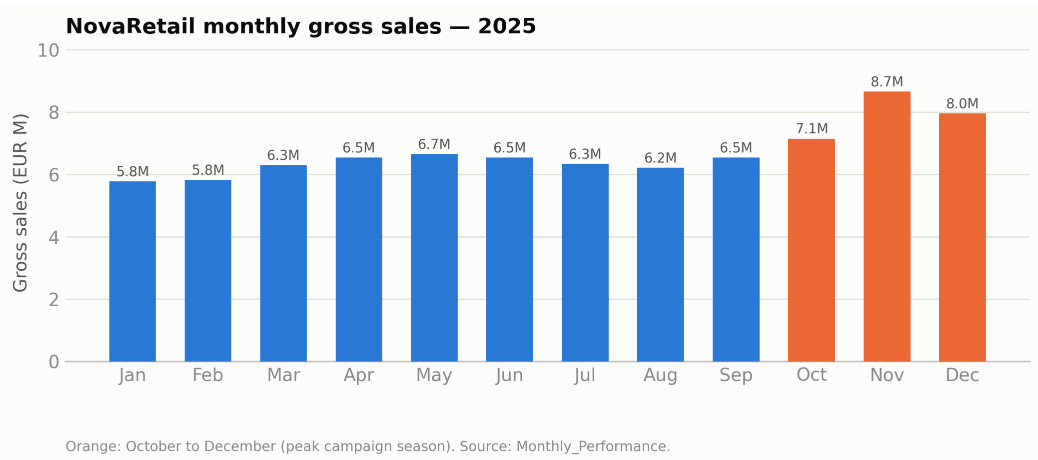

Figure 1. NovaRetail monthly gross sales in 2025; Q4, in orange, is the busiest commercial season of the year.

A simpler explanation? Purchasing power by region

Before isolating an internal cause, it is worth testing the simplest explanation: did these figures fall because purchasing power fell too? The UK & Ireland did in fact face the most adverse conditions of the six regions in Q4 2025 (inflation running near 3.0%, the highest of the group, real wages nearly stagnant, negative consumer confidence, according to the ONS and Eurostat), against 1.2% in Italy, where purchasing power actually grew. But this does not hold on its own: Iberia had similar inflation (close to 3.0%) and Fashion discount stayed stable there in Q4 (Figure 2). Weakened purchasing power is a plausible aggravating factor, not the sole cause, which reinforces the need to look at factors internal to NovaRetail, explored next.

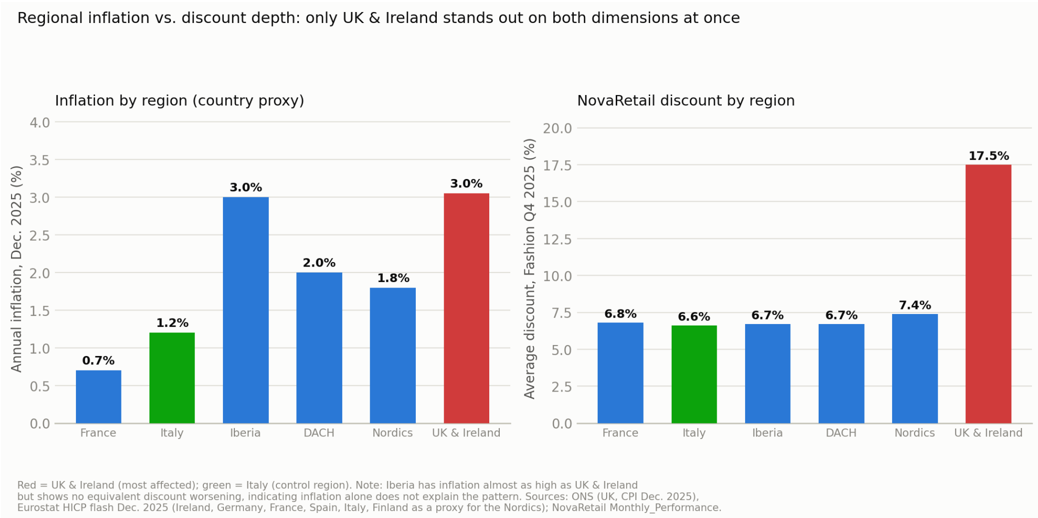

Figure 2. Inflation by region in Q4 2025 (country proxy) and NovaRetail's average Fashion discount; only the UK & Ireland stands out on both dimensions at once.

2. Business question and analytical question

Business question

Is Q4 promotional intensity (Black Friday and Christmas) in the Fashion category in the UK & Ireland destroying more value than it generates, through higher returns, margin compression and lower customer satisfaction? If so, should NovaRetail revisit the depth of that promotion before the next Q4, and should that decision be made centrally or by regional management?

Analytical question

Is there, in the 2025 data, a consistent relationship between discount, returns and gross margin in Fashion that is structurally different in the UK & Ireland compared with regions where the same pattern does not appear, such as Italy, once operational explanations and general seasonality effects are excluded?

Scoping the question to a specific combination of category, region and quarter, rather than asking generically what is wrong at NovaRetail, is what makes it possible to reach an actionable conclusion instead of a descriptive summary of the dataset.

3. GenAI as an analytical copilot: observations, hypotheses and validation

The exploration was carried out with the support of a Generative AI model, Claude, by Anthropic, as an analytical copilot, with actual Python code execution to calculate and verify every number cited; no value presented is a model estimate without an underlying calculation. The process followed the dataset's own principle: understand the structure of each table before cross-referencing, recompute composite indicators, and systematically separate observation from hypothesis. The appendix documents the prompts used and the points where an initial reading was corrected or rejected.

What the data shows: supported observations

- The relationship between discount and returns is far stronger in Fashion than in other categories: controlling for region and month (72 observations), the discount to return correlation is 0.90 and discount to margin is 0.97 negative; in the remaining categories it ranges between 0.02 negative and 0.42.

- Within Fashion, the Q4 pattern is not generalized: in five of the six regions discount stays stable between January and September and Q4; only the UK & Ireland spikes, from 7.3% to 17.5% (Figure 3).
- The most direct comparison is with Italy, where this pattern is practically absent: in the same quarter, Italy even improves slightly on all three indicators (Figure A and Table 1), on a trajectory opposite to the UK & Ireland's.
- The operational dimension does not explain the pattern: order fulfillment and on-time delivery stay stable (around 98% and 97%), pointing to a commercial rather than logistics cause.
- The three Fashion campaigns in the UK & Ireland in Q4 (€208 thousand invested) have ROAS of 0.55x, 0.31x and 0.74x, all below break-even, and none has an incrementality test.
- CSAT falls from 78.8 to 72.0, and return related inquiries to customer support more than double per month. Customer comments in Q4 point to sizing and fit issues as the reason for returns.

Plausible hypotheses, not yet fully proven

- The much deeper discount may have attracted exploratory buying (for example, ordering multiple sizes and returning the ones that do not fit), simultaneously inflating sales and returns.
- Q4 campaigns may have generated volume without setting the right customer expectations on sizing and fit, and the weak ROAS combined with margin compression suggests net value destruction; but a deliberate stock clearance cannot be ruled out.

What still requires validation

- The reason for each return is not available at the item level, so it is not possible to confirm that sizing and fit is the dominant reason at scale, rather than just in the cases cited in the sample.
- There is no history before 2025, so it cannot be ruled out that this is simply how the UK & Ireland has always run Black Friday, although the contrast with Italy makes that unlikely; the Q4 feedback sample is small (16 ratings), so the drop from 2.50 to 1.62 should be read with caution.

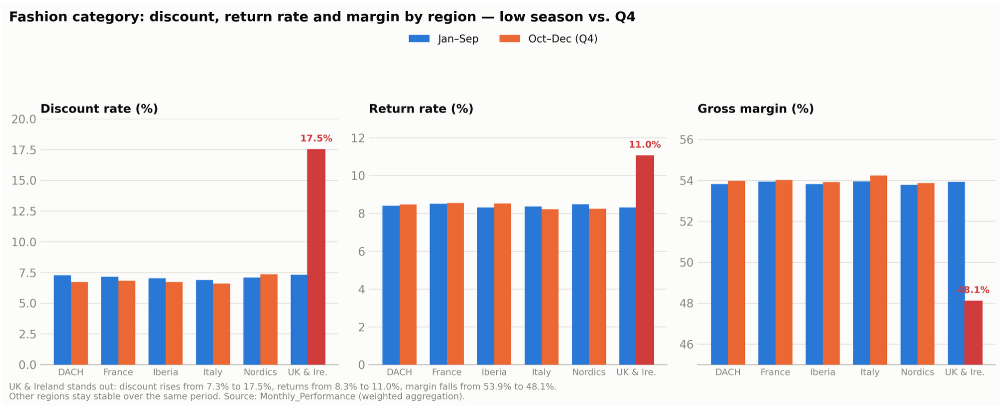

Figure 3. Fashion discount, returns and margin by region, January to September versus Q4. The UK & Ireland is the only outlier across all three dimensions.

Direct comparison: UK & Ireland versus Italy

The table below isolates the two regions that best illustrate the problem, in the same category and the same quarter.

Indicator (Fashion)	UK & Ireland Jan-Sep	UK & Ireland Q4	Italy Jan-Sep	Italy Q4
Discount rate	7.3%	**17.5%**	6.9%	**6.6%**
Return rate	8.3%	**11.0%**	8.4%	**8.2%**
Gross margin	53.9%	**48.1%**	53.9%	**54.2%**

Source: Monthly_Performance, weighted aggregation by units and value, not a simple average across rows, as recommended in the dataset's Data_Dictionary.

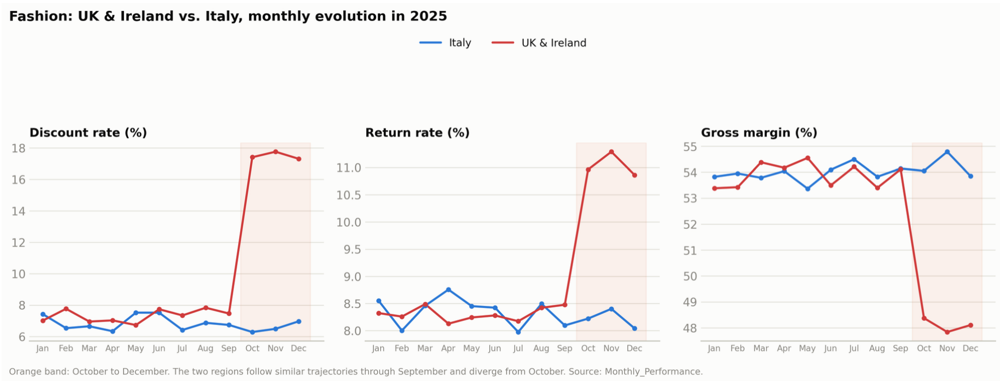

Figure 4. UK & Ireland versus Italy, monthly evolution in Fashion in 2025. The trajectories diverge precisely from October.

4. Key insights and supporting evidence

This section brings together cross-referenced signals from four independent sources (commercial, campaigns, customer service, feedback), all relating to Fashion, UK & Ireland, Q4. The convergence across sources is what sets this reading apart from an isolated correlation.

Area	Jan-Sep	Q4	Reading
Campaigns (ROAS and test)	n/a	**0.55x / 0.31x / 0.74x, no test**	3 Fashion campaigns in Q4, all below break-even (1x) and with no causal proof.
Customer service (CSAT, and return inquiries/month)	78.8 · 68	**72.0 · 150**	CSAT drops about 7 points and return inquiries more than double in Q4.
Customer feedback (1-5 scale)	2.50	**1.62**	Sharp drop; reduced sample in Q4, read with caution.
Operations (fill rate / on-time delivery)	about 98% / 97%	**about 98% / 97%**	Stable: the cause does not appear to be a stock shortage or delivery delay.

The insight is not that deep discounting generally causes returns (other regions discount without the same effect, and Italy even improves in the same quarter). It is more specific: Q4 discount in this region was calibrated at more than double the depth of the rest, which appears to cross a threshold beyond which customer behavior changes, with more exploratory buying and returns, without net revenue or satisfaction keeping pace with gross volume.

The campaign return angle

Fashion has the second worst average ROAS among categories (0.73x, ahead of only Personal Care), and 95% of the campaigns in the dataset (79 of 83) have no incrementality test, including the three Fashion campaigns in the UK & Ireland in Q4 (Figure 5). There is not yet enough causal evidence to automate investment decisions in this category.

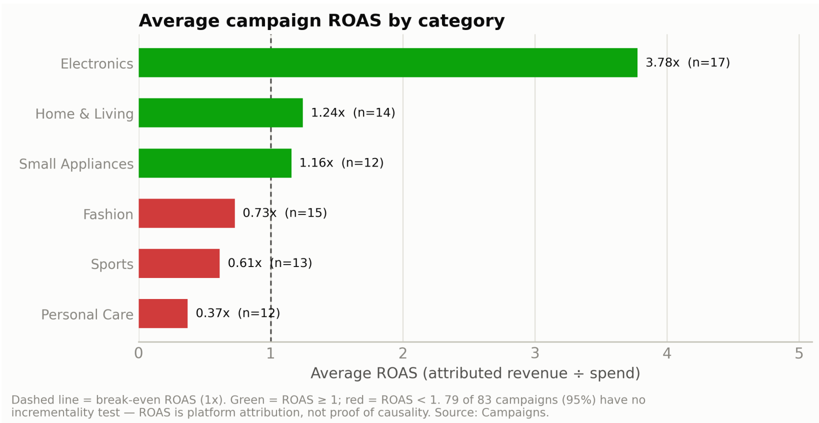

Figure 5. Average campaign ROAS by category. Fashion is among the categories with the worst return, with no incrementality test to confirm causality.

The customer angle

The three customer experience indicators (average rating, CSAT, return inquiries) move in the same direction: deterioration (Figure 6), reinforcing that this is not only a financial problem, but also a customer experience one.

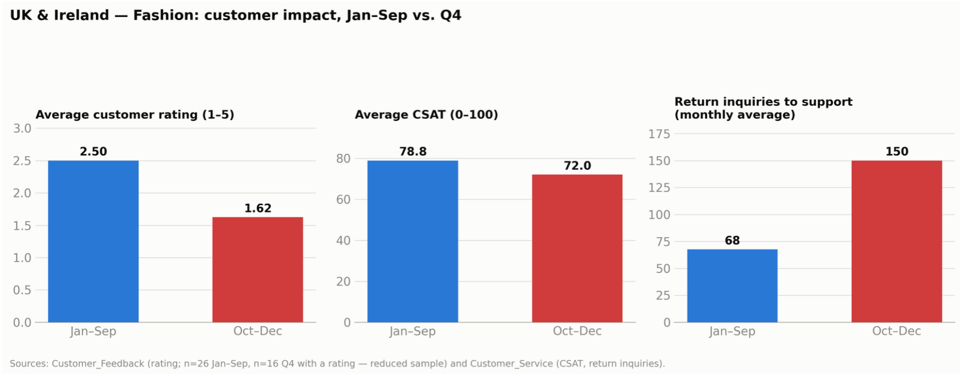

Figure 6. UK & Ireland, Fashion, customer impact, January to September versus Q4.

5. AI solution recommendation

Of the three possible responses, equipping users with a GenAI interface, building an automation, or creating a business level AI agent, the recommendation is option (a). A GenAI interface lets managers explore commercial, campaign, operations and customer service data in natural language, with answers grounded in the data and a clear distinction between observation, hypothesis and point to validate. It is better suited than an automation because the dominant cause is not yet known with confidence, and automating on top of an incomplete diagnosis risks fixing the wrong symptom. It is better suited than a business agent because the elements that would make one safe are still missing (return reason at the item level, multi year history, an incrementality test); without them, it risks becoming unstable. The interface also answers the need for fast explanations, without waiting for the next reporting cycle, serving managers across NovaRetail's six regions.

6. Expected value, risks, guardrails and KPIs

Expected value for the business

- Faster diagnostic cycle: managers no longer depend on an ad hoc request to the analytics team, and can get a first structured explanation in minutes, with the source cited.
- Margin protection during promotional periods: the estimated erosion in this combination alone is around €29 thousand in Q4 2025, and the real value is higher if the pattern repeats or exists in other combinations not yet identified, which the interface helps detect earlier.

Key risks

- Overconfidence or hallucination: the interface may present correlation as causation, generate plausible but ungrounded explanations, and propagate gaps (such as small samples) if it does not flag them.
- Inconsistent answers across users or sessions, which erode trust in the tool faster than any single mistake.
- Scope creep: users may start asking the interface to recommend or justify pricing and discount decisions, a function it has not been validated or governed for.

Guardrails and required human approvals

- The interface answers only from NovaRetail's own data, never from free generation, always shows its source, and labels each answer as observation, hypothesis or point to validate, flagging small samples.
- No automatic write access: the interface does not change prices, discounts or campaign budgets; any commercial action stays with the regional or category director.
- Mandatory human review before any generated number is cited externally or reported to executive leadership, with periodic sampling audits.

KPIs to measure impact

- Discount to return differential by region and category, measured monthly: measures whether promotional depth sits in a healthy range, with a target of bringing the UK & Ireland's Q4 spread closer to Italy's level.
- Gross margin protected during promotional periods (euros and percentage points): difference between Q4's actual margin and the prior period's baseline margin, tracked by region and category through the interface.
- Tool adoption and trust: queries per manager per month, and the percentage of answers validated as correct by users themselves, measuring reliability risk.

References

Office for National Statistics (ONS). *Quarterly economic commentary: October to December 2025.*

Office for National Statistics (ONS). *Consumer price inflation, UK: December 2025.*

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NIQ. *UK Consumer Confidence Increases Two Points in October to -17.*

KPMG. *Cautious Consumer Landscape Continues into 2026.*

Retail Times / Office for National Statistics (ONS). *Retail Sales Decline for Second Consecutive Month as Black Friday Fails to Entice Shoppers.*

Pound Sterling Live. *Pound to Euro Exchange Rate: 2025 Review.*

Central Statistics Office (CSO) Ireland. *Ireland 2025: The Year in Numbers.*

Trading Economics. *Ireland Inflation Rate.*

Il Sole 24 Ore. *In December, Inflation in Italy is Back on Track: +1.2% Year on Year.*

Eurostat. *Euro Area Annual Inflation Down to 2.0%, December 2025 Flash Estimate.*

Appendix. Evidence of GenAI use

This appendix concisely documents the prompts actually used during the analytical process and concrete examples of iteration and validation, including cases where an initial reading from the copilot was corrected or rejected after programmatic verification or cross referencing with additional data. Every numeric value cited in the body of the report was recalculated in Python, or verified against the cited external sources, never accepted from the GenAI model as the sole source of a number.

Moment	Prompt used (summarized)	Purpose / what it produced in the report
1. Initial prompt	*"Act as a senior data analyst preparing a report for management, to be delivered in Word and then PDF. Before that, run an exploratory analysis of the data using the uploaded Excel file and Python and Matplotlib, and use those charts to help build the report. The report should answer: identify a relevant business problem emerging from the case study and the dataset, explain why it matters, and translate it into a clear business question and a clear analytical question. Use Generative AI (GenAI) as an analytical copilot to explore the problem, clearly separating observations supported by the data, plausible hypotheses, and points that need further validation. Recommend the most suitable AI based response for this case: (a) equipping users with a Generative AI interface, (b) building an automation, or (c) creating a business level AI agent, and justify why the recommendation is better suited than the other two options. Define the expected business value, the main risks, the guardrails or human approvals required, and two to three KPIs to measure impact. For this assignment, I want to use option (a), equipping users with a Generative AI interface, since after the class this assignment is based on, I believe it is the most appropriate choice for working clearly with an LLM and, for the first time, building a project from scratch with well designed prompts."*	Defined the full scope of the work, the four questions the assignment required, and the initial, reasoned choice of option (a) as the AI response to develop.
2. Correction and refinement	*"The problem identification is great, but a few sections need fixing: in the AI recommendation paragraph, keep only the first paragraph, focused on option A alone. Remove every dash from the text, keeping normal grammar, more natural in tone but still professional. Since the problem is concentrated in one region, add a more direct comparison with the locations where the same problem is not as pronounced. The report should be 6 pages long, leaving room at the end for the prompts used, starting on page 8."*	Simplified the recommendation section to a single paragraph, removed dashes throughout the text, introduced the direct UK & Ireland versus Italy comparison, and fixed both the body length and where the prompts appendix would sit.
3. Iteration: regional purchasing power	*"Based on the problem framing (NovaRetail, six regions, €80.5 million in gross sales in 2025, 34.5% average gross margin, 5.5% overall return rate): show a comparison between the decline in these figures and purchasing power by region, factoring in economic and political conditions and each country's economic strength."*	Prompted research into real macroeconomic indicators (inflation, wages, consumer confidence) by region for Q4 2025, and the resulting purchasing power section in the report, including the Iberia counterpoint.
4. Iteration: executive summary chart	*"In the executive summary, it would help to have a chart comparing the two locations, UK & Ireland and Italy. Build a bar chart in Python comparing the percentages."*	Produced the bar chart (Figure A) in the executive summary, comparing discount, returns and margin between the two regions in Q4.
5. Visual identity	*"Given how much visual presentation matters for a piece of work like this, present three visual identity concepts for the company, including a logo and a banner for company documentation and for this report."*	Generated three visual identity concepts (logo and banner); the chosen concept, "Corporate Blue Nova," was applied to the report's cover and footer.

Moment	Prompt used (summarized)	Purpose / what it produced in the report
6. Restructuring the problem framing section	*"This part should be removed from the report, and in that same place, frame the analysis above, as I asked for in this prompt: Building on this topic: "1. Problem framing. NovaRetail is a European retail group operating across six management regions, DACH, France, Iberia, Italy, UK & Ireland and Nordics, six product categories and three sales channels. In 2025 the company generated €80.5 million in gross sales, with an average gross margin of 34.5% and an overall return rate of 5.5%. The executive team no longer believes the company faces a single, company wide problem: instead, it suspects that different regions and categories face distinct combinations of commercial, operational and customer facing problems, and that is exactly the hypothesis the data confirms." Please show me a comparison between the decline in these figures and purchasing power by region. In other words, did these regions' figures fall because purchasing power fell too? Economic and political factors, and each country's economic strength, can come into play here."*	Prompted the removal of the block of executive quotes that had been in the problem framing section, replacing it in the same place with the regional purchasing power analysis (text and Figure 2).